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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



TEMPORO MANDIBULAR JOINT DISORDERS

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Outline:

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- ❖ Overview of TMJ anatomy
- ❖ Disorders of TMJ
 - ✓ Hypermobility of TMJ
 - Acute dislocation
 - Chronic recurrent dislocation
 - ✓ Myofacial pain dysfunction syndrome
 - ✓ Hypomobility of TMJ- Ankylosis
- ❖ Non surgical and surgical management of TMJ disorders



ANATOMY OF TMJ

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- Discussion

Anatomy of TMJ

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- The temporomandibular joint (TMJ) is a complex structure composed of the temporal bone, the mandible, the articular disk, several ligaments, and numerous associated muscles.
- The TMJ is a specialized joint that can be classified by anatomic type as well as by function.

Anatomically

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- ✓ From an anatomic standpoint, the TMJ is classified as a *diarthrodial joint*, which is a discontinuous articulation of two bones permitting freedom of movement that is dictated by associated muscles and limited by ligaments.
- ✓ The TMJ is also a synovial joint, lined on its inner aspect by a synovial membrane, which secretes synovial fluid.
- ✓ The fluid acts as a joint lubricant and supplies the metabolic and nutritional needs of the non vascularized internal joint structures.

Cont.

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- From a functional standpoint, the TMJ is classified as a compound joint, composed of four articulating surfaces: the articular facets of the temporal bone (1) and of the mandibular condyle (2) and the superior (3) and inferior (4) surfaces of the articular disk.
- The articular disk divides the joint into two compartments.
- The lower compartment permits hinge motion or rotation and hence is termed ginglymoid.
- The superior compartment permits sliding (or translatory) movements and is therefore called arthrodial.
- Hence the temporomandibular joint as a whole can be termed ginglymoarthrodial.

Articular cartilage

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- The articular surfaces of both the temporal bone and the condyle are covered with dense articular fibrocartilage, termed as articular cartilage.
- Articular cartilage is composed of chondrocytes and an intercellular matrix of collagen fibers, water, and a nonfibrous filler material, termed *ground substance*.
- This Fibrocartilage covering layer has the capacity to regenerate and to remodel under functional stresses.

Articular disc

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- The articular disk is composed of a dense fibrous connective tissue, and it is non vascularized and non innervated; this is an adaptation that allows the disk to resist pressure.
- Anatomically, the disk can be divided into three general regions as viewed from the lateral perspective: the anterior band, the central intermediate zone, and the posterior band.
- The thickness of the disk appears to be correlated with the prominence of the eminence, such that proportionally, the anterior band (3, thickest), intermediate zone (1, thinnest), and posterior band (2, middle thickness) have relative thicknesses.

Cont.

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- Posteriorly, the articular disk blends with a highly vascular, highly innervated structure known as the *retrodiskal tissues*.
- Anatomically, the retrodiskal tissues are referred to as the *bilaminar zone* (*superior and inferior retrodiskal laminae*), which is involved in the production of synovial fluid.

Ligaments

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- Ligaments associated with the TMJ are composed of collagen and act predominantly as restraints to motion of the condyle and the disk.
- Three ligaments—collateral, capsular, and temporomandibular ligaments—are considered functional ligaments because they serve as major anatomic components of the joints.
- Two other ligaments—sphenomandibular and stylomandibular—are considered accessory ligaments because, although they are attached to osseous structures at some distance from the joints, they serve to some degree as passive restraints on mandibular motion.

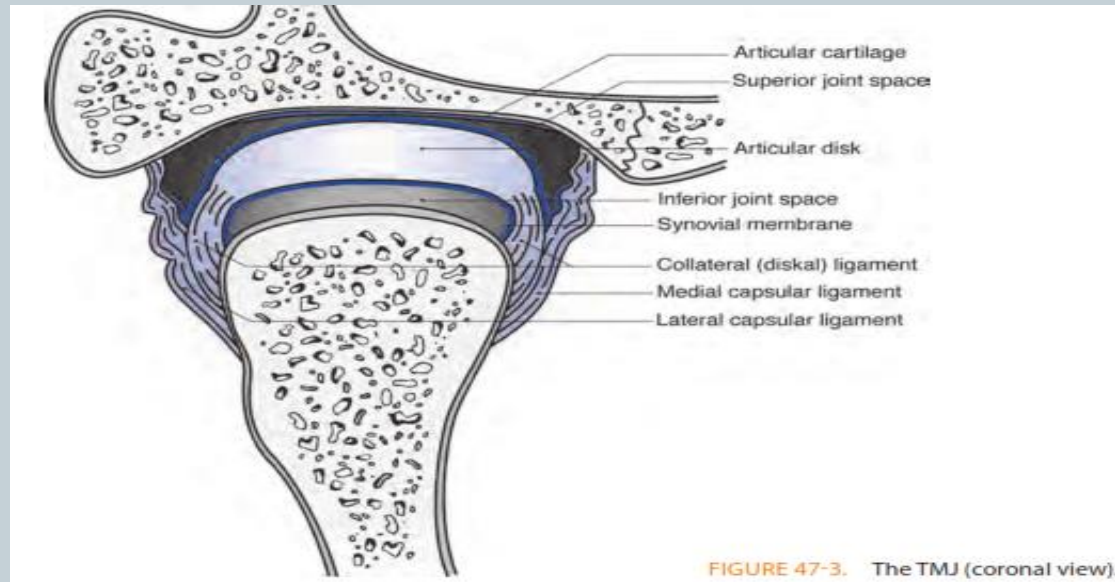
Muscles of TMJ

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- Generally 12 pairs of muscles act on TMJ.

But primarily,

- Temporalis ms.
- Masseter ms.
- Medial Pterygoid ms.
- Lateral Pterygoid ms.



Neurovascular supply

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- The vascular supply of the TMJ arises primarily from branches of the superficial temporal and maxillary arteries posteriorly and the masseteric artery anteriorly.
- The nerve supply to the TMJ is predominantly from branches of the auriculotemporal nerve with anterior contributions from the masseteric nerve and the posterior deep temporal nerve.

Pathology of TMJ

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- The demand for treatment of temporomandibular joint dysfunction (TMJD) is well known.
- Most studies estimate the prevalence of clinically significant TMJ-related jaw pain to be at least 5% of the general population.
- Approximately 2% of the general population seeks treatment for a TMJ-related symptom.

Classification

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- TMJD may be the result of muscular hyperfunction or parafunction and/or underlying primary or secondary degenerative changes within the joint.
- Classification of TMJD is separated into non articular and articular categories and has been eloquently described by de Bont and colleagues.

TABLE 47-2. General Classification of Synovial Disorders

Articular Disorders

Noninflammatory Arthropathies

Primary idiopathic osteoarthritis

Secondary osteoarthritis (trauma, prior surgery, avascular necrosis)

Mechanical derangements

Bone and cartilage disorders, with articular manifestations

Inflammatory Arthropathies

Synovitis

Capsulitis

Rheumatoid arthritis

Juvenile rheumatoid arthritis

Seronegative polyarthritis

Ankylosing spondylitis

Psoriatic arthritis

Reactive arthritis (bacterial, viral, fungal)

Growth Disturbances

Non-neoplastic: developmental (hyperplasia, hypoplasia, dysplasia)

Non-neoplastic: acquired (condylolysis)

Neoplasms

Pseudotumors (synovial chondromatosis)

Benign (chondroma, osteotoma)

Malignant (primary, metastatic)

Diffuse Connective Tissue Disorders

Miscellaneous Articular Disorders

Nonarticular Disorders

Muscular Disorders

Muscle spasm (strain)

Myofascial pain and dysfunction

Fibromyalgia

Myotonic dystrophies

Myositis ossificans progressiva

Growth Disturbances

Nonarticular TMJD

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- Nonarticular TMJ disorders most commonly manifest as masticatory muscle dysfunction.
- Approximately one half or more of all TMJDs are forms of masticatory myalgia.
- They include such conditions such as acute muscle strain and spasm, myofascial pain and dysfunction (MPD), chronic conditions such as fibromyalgia, and less commonly, myotonic dystrophies and myositis ossificans.

MPDS

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- **DISCUSSION?**

MPD

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- MPD is most commonly related to masseter or temporalis muscle spasm.
- In addition, it can involve the pterygoids or any combination of the supramandibular or inframandibular muscle groups.
- Parafunctional habits such as bruxism and jaw clenching are thought to be the main contributors to MPD and have also been found to be causative in acute closed-lock conditions.

Pathophysiology

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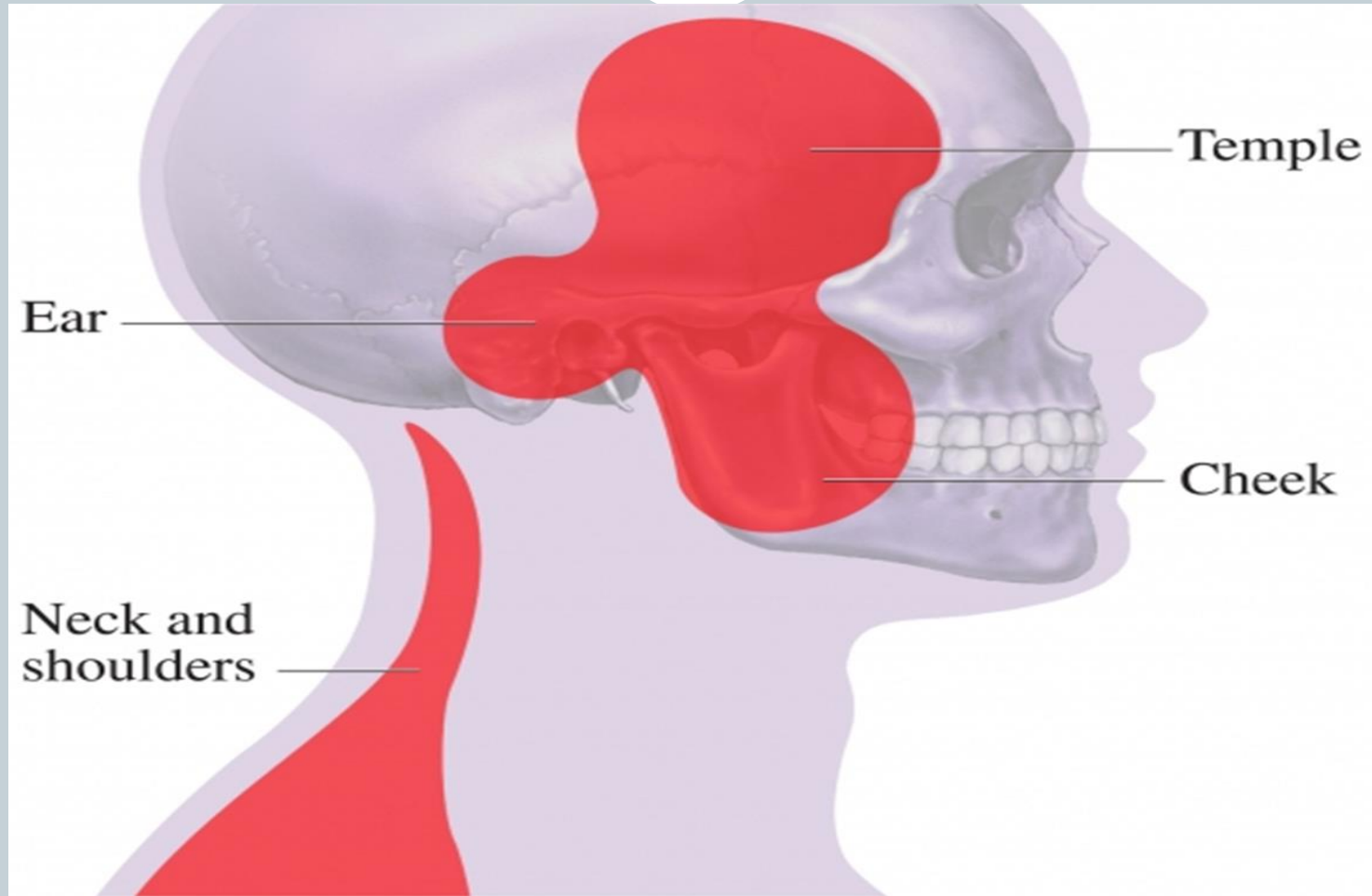
- Muscle injury causes increased tone of musculature which leads to muscle fatigue & accumulation of metabolic byproducts causing pain.

Signs & symptoms

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- Pain or discomfort anywhere in head & neck region.
- Limitation of motion of jaw.
 - normal vertical range of motion is 40 to 50 mm.
 - normal lateral range of motion is 10mm.
 - normal range of protrusive movement is 10mm.
- Joint noises-clicking.

- Headache
- Burning or tingling sensation
- Tenderness
- Ear pain w/o infections
- Neck and or facial pain
- Grinding teeth at night
- Pain that worsens with stress
- Pain with opening of your mouth
- Pain while chewing



RX



- There are various treatment modalities for MPD.
- Such treatments include occlusal adjustments (for gross discrepancies), night-guard appliances, NSAIDs, muscle relaxants, and physical therapy.
- These treatment modalities, alone or in combination, remain the standard of care for the treatment of nonarticular TMJD, particularly MPD.



- Fibromyalgia is a systemic condition marked by poor sleep, generalized pain with absence of localization to joints, and a history of somatization in other organ systems such as irritable bowel syndrome and headache.
- It is typically observed in an older population than MPD and has a female predilection.

- Rarely, other nonarticular conditions such as myotonic dystrophy and myositis ossificans progressiva can lead to significant loss of function and pain in the TMJ region.
- Myotonic dystrophy is a dominantly inherited multisystem disorder that may affect facial muscles in fully developed disease states.
- This condition may contribute to atrophy and fibrosis of the supramandibular and inframandibular musculature.

- Myositis ossificans progressiva is a rare condition resulting in fibrosis of soft tissues after apparent minor trauma.
- This condition has been reported to affect TMJ function after local trauma, including surgery, and can result in significant loss of mandibular range of motion, trismus, and pain.

ARTICULAR DISORDERS

- Noninflammatory articular disorders of the TMJ, the most common of which is osteoarthritis, are often idiopathic.
- Osteoarthritis can manifest as chondromalacia (softening of the cartilage), temporary or permanent disk displacement, degenerative changes within bone and cartilage often with osteophyte formation and remodeling, fibrosis, or any combination of these.

Cont.

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- Noninflammatory articular disorders may also be secondary to trauma, infection, previous surgery, crystal deposition disorders (gout), avascular necrosis, or structural damage to joint cartilage resulting in disk displacement and/or perforation.

Discussion & Reading assignment

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- TMJ disk displacement has been categorized through a widely accepted staging system by Wilkes, using such criteria as severity of displacement and chronicity.(reading assignment)
- TMJ disc displacement with and without reduction?

Inflammatory

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- Inflammatory arthropathies are primarily due to such conditions as rheumatoid arthritis, juvenile rheumatoid arthritis, ankylosing spondylitis, psoriatic arthritis, or arthritis resulting from infectious cause.
- Secondary causes of inflammatory arthropathies include synovitis, capsulitis, traumatic arthritis, or acute inflamed crystal-induced arthritis, such as gout.

- History
- Clinical examination
- Radiographs
- Lab investigation?

- Treatment of patients with internal derangement of the TMJ typically begins with nonsurgical treatment modalities.
- Bite appliance therapy, diet modifications, NSAIDS, muscle relaxants, moist heat or ice, and physical therapy have been found to be efficacious.
- Surgical intervention is typically employed only after failure of nonsurgical treatment objectives.

Surgical

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- A variety of surgical treatment modalities have been used in the treatment of articular TMJD.
- Arthrocentesis and TMJ arthroscopy have been found to be minimally invasive effective treatments for articular TMJD by decreasing pain and increasing mandibular range of motion.

Cont.

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- Arthrocentesis and arthroscopy have also been reported to be useful in severe, often sudden-onset, closed-lock disease due to an anchored or “stuck disk” phenomenon.
- This proposed phenomenon involves the disk becoming adherent to the glenoid fossa through increased intra-articular friction, with or without the formation of adhesions within the joint.

Infections of TMJ

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- Infections of the TMJ are not common.
- Prompt diagnostic and therapeutic intervention is required when an infection of the TMJ is suspected because joint distention is usually painful and permanent changes in joint function can occur.

Clinical features

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- On examination, patients usually exhibit a posterior open bite on the ipsilateral side as a result of the increased joint fluid.
- The patient will also maintain a posture toward the contralateral side.
- The surface overlying an infected joint is often warm, and fluctuance is occasionally felt.

Etiologies

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- The bacteria causing an infected joint are usually spread through a hematogenous route.
- The synovium is vascular and lacks a basement membrane, which permits bloodborne bacteria to gain access to the joint space.
- Other direct injury, from infected third molar.

- History and clinical examination plus
- The synovial fluid should be Gram-stained and cultured for both aerobic and anaerobic bacteria.
- The aspirate from the infected joint should be submitted for white blood cell (WBC) count and differential and examined for the presence of crystals and fibrinogen.
- Fibrinogen is usually present in the synovial fluid of acutely infected joints.

Microorganisms

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- Staphylococcus aureus and N. gonorrhoeae
- Haemophilus influenzae
- Borrelia burgdorferi

- Ampicillin and sulbactam
- One-month course of doxycycline or amoxicillin
- Clindamycin
- Tobramycin
- Arthrocentesis and active exercise after treatment of the infection.

Dislocation

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- DISCUSSION
- WHAT IS MEAN BY DISLOCATION
- CLASSIFICATION AND TYPES OF DISLOCATION
- CAUSES AND PREDISPOSING FACTORS
- MENTION THE CLINICAL FEATURES OF DISLOCATION
- MGT

Dislocation

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- Condylar head is displaced out of glenoid fossa, much anteriorly beyond articular eminence, but still remains within the capsule of the joint
- unilateral or bilateral.
- Anterior mandibular dislocation can be classified as
 1. Acute
 2. Chronic recurrent (habitual) subluxation
 3. Long-standing.

Cont..

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Causes of Acute Dislocation

- ❑ Extrinsic forces or iatrogenic causes
- ❑ Intrinsic or self-induced forces

Cont..

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- Anterior acute dislocation of the condyle occurs, in which the normal anatomic relationships within the joint have been completely disrupted, with the condyle fixed anterior to the articular eminence.
- Relocation of the condyle in its normal position in the glenoid fossa does not occur voluntarily.
- It can occur as a single acute event or as chronic recurrent episodes

Extrinsic or iatrogenic causes

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Acute dislocation is common and can be brought about by

- Blow on the chin, while mouth is open.
- During general anaesthesia
- During dental extraction.

Intrinsic or self-induced forces

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- Excessive yawning
- Singing loudly
- Blowing wind instruments
- Laughing loudly or opening mouth too wide for eating

Predisposing factors

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- Laxity of ligaments, capsule and abnormality of skeletal form.
- Flattened eminence and shallow fossa
- systemic diseases like Parkinson's disease and epilepsy.
- Use of antipsychotic drugs may cause extrapyramidal reactions and dislocation.

Clinical picture

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- It may be unilateral or bilateral.
- History of the patient may be diagnostic
- ❖ **Unilateral acute dislocation**
 - Difficulty in mastication and swallowing and Speaking
 - Profuse drooling of saliva can be present
 - A deviation of the chin toward contralateral side is seen.
 - Lateral cross bite and open bite on the contralateral side.
 - Mouth is partly open and the affected condyle cannot be palpable.

Bilateral acute dislocation

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- ✓ Associated with pain?, inability to close the mouth, difficulty in speech, excessive salivation and protruding chin.
- ✓ The mandible is postured forward and movements are restricted.
- ✓ There is a gagging of the molar teeth with the presence of anterior open bite.
- ✓ Hollowness can be felt in both the preauricular regions.

Management

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- ❖ Initially reduce tension, anxiety and muscle spasm
 - (i) reassuring the patient,
 - (ii) tranquilizer or sedative drugs,
 - (iii) pressure and massage to the area
- ❖ Manipulation
 - acute dislocation can be reduced by
 - 1. Manipulation without any form of anesthesia.
 - 2. Manipulation with local anesthesia.
 - 3. Manipulation under general anaesthesia/sedation with muscle relaxants.

Reduction

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- ✓ Reading assignment on the various suggested reduction techniques?
- ✓ Stand in front of the patient
- ✓ Grasp the mandible with both the hands, cover thumbs with gauze to prevent injury during manipulation, as sudden reduction can take place trapping the thumbs of the operator by the teeth.
- ✓ The thumbs are placed on the occlusal surfaces of the lower molars and fingertips are placed below the chin.
- ✓ Exert full body pressure and give downward pressure on the posterior teeth and backward and upward pressure to the jaw.

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Cont..

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- ❖ After reduction procedure,
 - Advice for oral opening restriction.
 - Immobilization , by giving barrel bandage to the patient for the period of 10 to 14 days
 - Semisolid diet.
 - NSAID for 3 to 5 days.
 - The patient is warned to avoid excessive oral opening and support the chin, while yawning in future.

Cont.d

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- ‘Long-standing’ is applicable to those cases, which remain unusually dislocated for longer than one month
- Follows extraction of teeth or tonsillectomy under general anaesthesia
- Long-standing acute dislocation, which does not respond to the above procedure can be reduced by administering general anaesthesia.
- If manual reduction fails, then open surgical procedure.

Chronic Recurrent or Habitual Dislocation or Subluxation

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- Repeated episodes of dislocation, where there is abnormal anterior excursion of the condyles beyond the articular eminence, but the patient is able to manipulate it back into normal position.
- Condylar head moves, unassisted, forward and backward over the articular eminence.
- This recurrent, incomplete, self-reducing, habitual dislocation is termed as hypermobility or chronic subluxation of the TMJ.

- Triad of ligamentous and capsular flaccidity?
- Eminential erosion and flattening
- Shallow glenoid fossa & trauma is well-recognized factors.
- Medications?

Medications that can cause dislocation:

Antipsychotic drugs: The use of antipsychotic medications can induce extrapyramidal reactions (involuntary muscle movements or spasms) that may lead to TMJ dislocation

Medications and injections used for treatment: When conservative management fails, clinicians may attempt to restrict the excessive joint movement by injecting specific substances into the joint space or surrounding musculature:

Sclerosing solutions: These are injected into the joint space to intentionally cause fibrosis (scarring) and tighten the capsule. Sodium psyllate historically provided the best results but is no longer available

. Sodium morrhuate has been tried but failed to produce good results, and sodium tetradecyl sulfate can be used with caution due to the risk of allergic or anaphylactic reactions

. Other sclerosing agents that have been utilized include alcohol and Rivanol-aethedicaïne

Autologous Blood and PRP: Injecting the patient's own autologous blood or platelet-rich plasma (PRP) into the joint and capsule to bring about stabilizing fibrosis is a technique that is becoming increasingly popular

Botulinum toxin (Botox): Intramuscular injections of Botulinum toxin type A directly into the lateral pterygoid muscle have been used to reduce the muscular pull that contributes to the dislocation, although the stabilizing results are often short-lived

Management?

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- ❖ Intermaxillary fixation or limiting the oral opening by giving elastics
- ❖ Total immobilization of the jaw for the period of 3 to 4 weeks gives rest to the joint.
- ❖ Kept on liquid diet.
- ❖ Other mgt options
- ❖ Sclerosing solution injections
- ❖ Surgical?

Hypomobility of Temporomandibular Joint

ANKYLOSIS OF TMJ

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- ANKYLOSIS A GREEK WORD MEANS STIFF JOINT.
- Hypomobility to immobility of the joint can lead to inability to open the mouth from partial to complete.

Table 22.1: Etiopathology of ankylosis of TMJ

- | | |
|---|---|
| <ul style="list-style-type: none">• Trauma• Infections• Inflammation• Rare causes• Systemic diseases• Other causes | <ul style="list-style-type: none">• Congenital• At birth, forceps delivery• Hemarthrosis (Direct/indirect trauma)• Condylar fractures<ul style="list-style-type: none">— intracapsular— extracapsular• Glenoid fossa fracture (rare)• Otitis media• Parotitis• Tonsillitis• Furuncle• Abscess around the joint• Osteomyelitis of the jaw• Actinomycosis• Rheumatoid arthritis• Osteoarthritis• Septic arthritis—hematogenous spread• Polyarthrititis• Measles• Smallpox• Scarlet fever• Typhoid• Gonococcal arthritis• Scleroderma• Beriberi• Marie-Strümpell disease• Ankylosing spondylitis• Bifid condyle• Prolonged trismus• Prolonged immobilization• Unknown• Burns |
|---|---|

Classification

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- Ankylosis may be classified according to location (intra-articular vs extraarticular)
- Type of tissue involved (bony, fibrous, or mixed),
- The extent of fusion (complete vs incomplete).
- The definite cause of ankylosis of TMJ is unknown.
- Two main factors predisposing to the ankylosis are trauma and infection in or around the joint region.

Pathogenesis

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- ❖ Trauma will bring about extravasation of blood into the joint space called haemarthrosis.
- ❖ This predisposes to calcification and obliteration of a joint space, where immobility of the joint is maintained over a prolonged period.
- ❖ Many times initial fibrous bands lead toward bony consolidation to ossification.

- Relatively easy and is dependent more upon careful clinical examination, rather than the diagnostic tests.
- Restricted oral opening or nil oral opening
- difficulty in mastication
- Difficulty in protrusive movements

Clinical manifestations

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- Clinical manifestation vary according to :
 - a. Severity of ankylosis
 - b. Time of onset of ankylosis
 - c. Duration
 1. Early joint involvement-less than 15 years: severe facial deformity and loss of function
 2. Late joint involvement after the age of 15 years: facial deformity marginal or nil. But ,functional loss severe

Unilateral ankylosis

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- Deviation of the mandible and chin on the affected side
- Hypoplastic mandible
- Roundness and fullness of the face on the affected side.
- The appearance of the flatness and elongation on the unaffected side.
- Well defined antegonial notch.
- Cross bite
- Class II angles malocclusion on the affected side plus unilateral posterior cross bite on the ipsilateral side

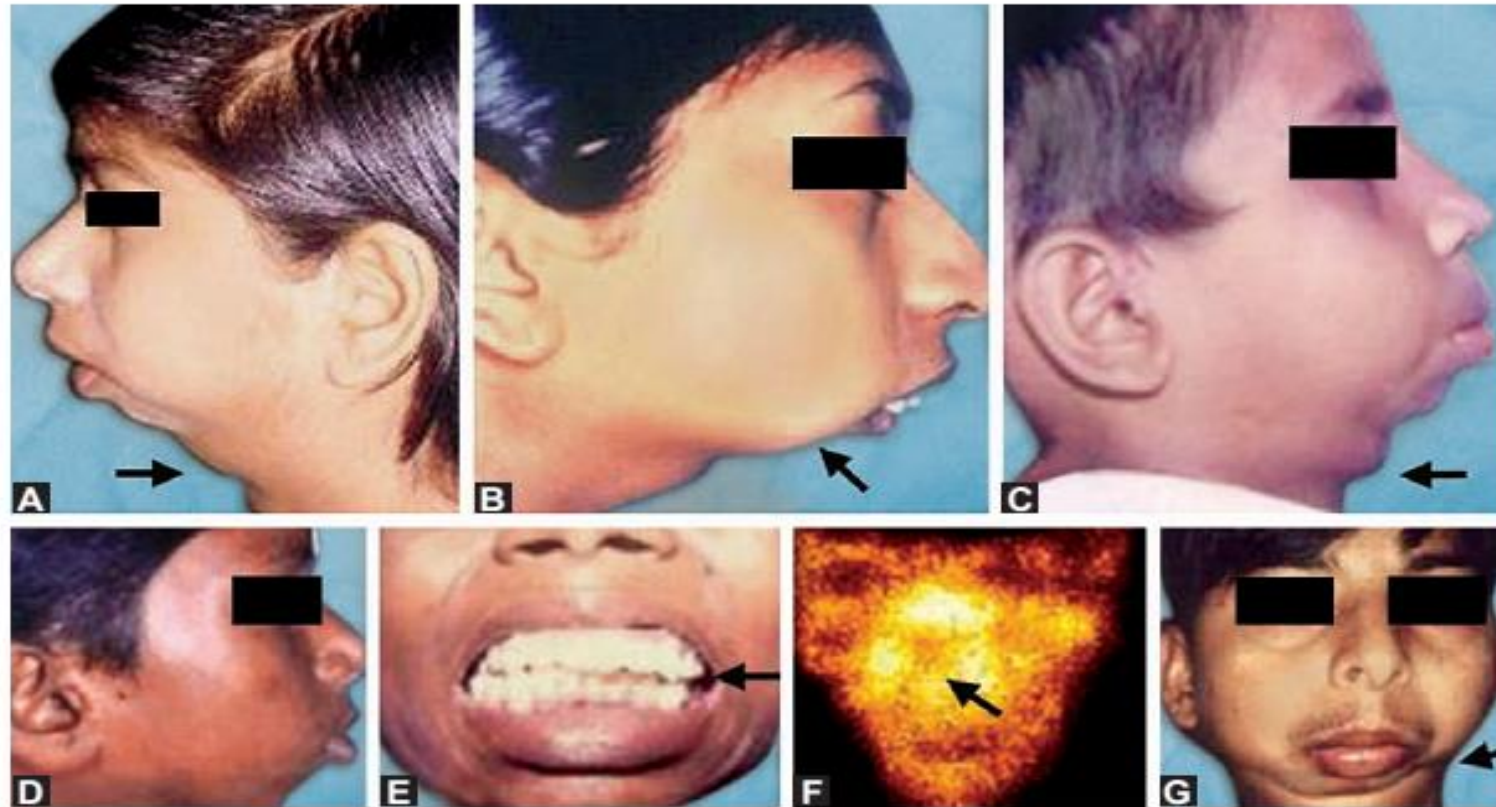


Fig. 22.3: Unilateral ankylosis: Facial and dental features

Bilateral

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- Typical bird face deformity
- Protrusive upper incisor with anterior open bite
- Bilateral well defined antegonial notch
- Class II angles malocclusion
- Decreased neck chin angle



Figs 22.4A to G: (A to C) Bilateral ankylosis of TM joint showing typical "Bird Face" and dental features, (D and E) Facial profile and nil oral opening in bilateral TMJ ankylosis, (F) Scintigraphy showing increased uptake in TMJ areas, indicating active reankylosis process, (G) Severe micrognathia in bilateral TM joint ankylosis case

Radiograph

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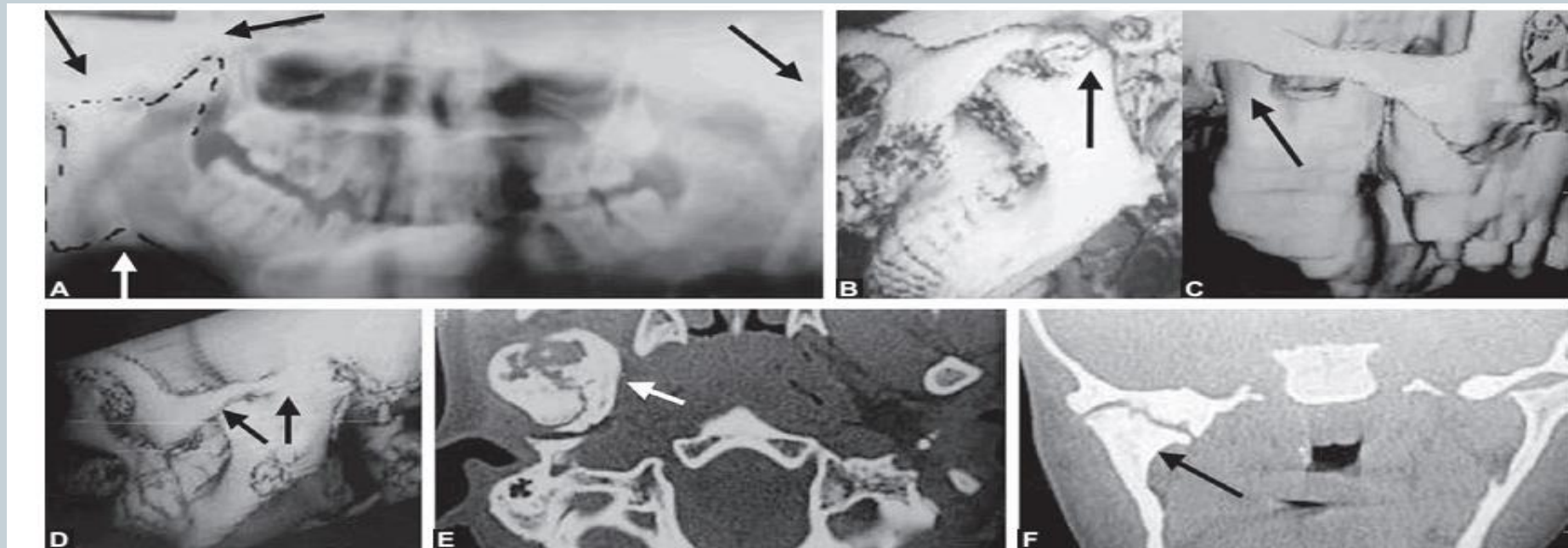
- Radiographic assessment is critical in evaluating and treating patients with hypomobility disorders
- Discussion ?

Radiograph

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- Radiographic assessment is critical in evaluating and treating patients with hypomobility disorders
- OPG:-presence of antegonial notch can be detected.
- Lateral oblique view:-gives anteroposterior dimension of condylar mass. Elongation of coronoid process can be seen.
- Posteroanterior radiograph:-highlight asymmetry in unilateral cases.
- computed tomography (CT) scans (including axial, coronal, and sagittal views with 3D) is helpful in fully defining the extent of ankylosis as well as the relationship of the ankylotic mass to important anatomic structures, especially at the skull base.

- Magnetic resonance imaging (MRI) has had a great impact on TMJ evaluation, especially regarding the delineation of meniscal position.
- Diagnosis of fibrous ankylosis is possible with the use of MRI, but the CT scan is superior in demonstrating bony pathology



Figs 22.6A to F: (A) OPG bilateral ankylosis, (B) Unilateral ankylosis-3D scan, (C) Normal TMJ in 3D scan, (D) (L) TMJ ankylosis 3D scan showing bony fusion and elongated coronoid process, (E) Axial scan—bony ankylosis, (F) Coronal scan—bony ankylosis

Radiographic findings

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Fibrous ankylosis:-

- reduced joint space
- hazy appearance
- normal anatomy can be appreciated

Bony ankylosis:-

- complete obliteration of joint space.
- deformed condylar head
- elongation of coronoid process

Sequelae of Untreated Ankylosis

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- ✓ Normal facial growth and development affected.
- ✓ Speech impairment.
- ✓ Nutritional impairment.
- ✓ Respiratory distress (tongue fall in sleep), especially, in bilateral involvement with severe micrognathia.
- ✓ Malocclusion.
- ✓ Poor oral hygiene.
- ✓ Multiple carious and impacted teeth

Aims and Objectives of Surgery

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1. Release of ankylosed mass and creation of a gap to mobilize the joint.
2. Creation of a functional joint.
 - ✓ To improve patient's nutrition.
 - ✓ To improve patient's oral hygiene.
 - ✓ To carry out necessary dental treatment.
3. To reconstruct the joint and restore the vertical height of the ramus.
4. To prevent recurrence

Management of TMJ Ankylosis

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Surgical.

- ✓ Early surgical correction of the ankylosed joint is highly desirable, if satisfactory function is to be regained.
- ✓ Surgical strategy adopted depends on the following:
 - a. Age of onset of ankylosis.
 - b. Extent of ankylosis.
 - c. Whether there is unilateral or bilateral involvement.
 - d. Associated facial deformity

SURGICAL

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❖ Condylectomy:-

- indication:-fibrous ankylosis in which there is no much deformity of condyle.

❖ Gap arthroplasty:-

- indication:- bony ankylosis, thick area of bone obliterates entire joint ,sigmoid notch & coronoid process.

❖ Interpositional arthroplasty

- Involves creation of a gap & insertion of a barrier b/w the cut bony surfaces to minimize recurrence

❖ Artificial replacement of joint

- Prefabricated condylar prosthesis made of steel or vitallium can be used.

Surgical approaches to TMJ

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1. Submandibular (Risdon's) approach

2. Postramal (Hind's) approach

3. Postauricular approach

4. Endaural approach

5. Preauricular approach

Dingman's

Blair's

Thoma's

Popowich's modification of Al-Kayat and Bramley's

6. Hemicoronal approach

7. Coronal or bicoronal approach

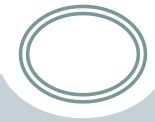
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The End



DEC, 2025

THANK YOU